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# GROUNDWATER POTENTIAL MAPPING USING AHP AND MACHINE LEARNING IN THE UPPER SUBARNAREKHA RIVER BASIN, INDIA

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## ABSTRACT

Groundwater is a key water source, and identifying high-potential regions is essential for effective management. Analytical Hierarchy Process (AHP) and Random Forest (RF) model have been used to map Groundwater Potential Zones (GWPZs) in upper Subarnarekha River Basin (SRB), India. Ten factors influencing groundwater occurrence were selected for inter-thematic correlation analysis. The model's performance has been assessed using Receiver Operating Characteristics (ROC). AHP shows approximately 15% of the area has good Ground Water Potential (GWP), primarily located in the south and southeast region. Additionally, around 57% of the area demonstrated moderate GWP in the central part of the basin, whereas 24% of the area shows poor GWP in the northwestern regions. ROC revealed that AHP and RF models achieved success rates of 71%, and 73%, respectively. Due to the limited availability of well data (126 data points across a 14,037 km² study area), the Machine Learning (ML) model struggled to accurately delineate groundwater potential zones. In contrast, the AHP-based weighted overlay analysis proved more effective. The findings can support watershed management efforts and guide the selection of optimal well locations in future projects.

***Index Terms—*** *Groundwater Potential (GWP), Analytic Hierarchy Process (AHP), Machine Learning (ML), Subarnarekha River Basin (SRB).*

## 1. INTRODUCTION

Water is a vital and dynamic resource essential for life and human activities. Rapid population growth has significantly impacted its availability and quality. Consequently, groundwater has become a key alternative water source, crucial for socio-economic development. Despite regional variations in quality and quantity, it meets essential needs like drinking water, irrigation, industry, and livestock support [1]. Excessive groundwater use has depleted this vital resource in many regions, making freshwater availability a growing challenge due to high demand for domestic, agricultural, and industrial purposes [2]. As a result, mapping groundwater potential zones (GWPZs) has become a crucial aspect of watershed management systems.

A number of variables, including lithology, geography, geological formations, climate, and soil type, affect the availability of groundwater resources. However, its mobility is mainly based on the porosity, permeability, transmissibility, and storage capacity of rocks [3]. Hence, it can be challenging to accurately pinpoint areas of high groundwater yield. Though effective in providing detailed information about hydrogeological conditions, traditional methods such as sample drilling and pumping tests can be expensive and time-consuming [4]. Integrating remote sensing with GIS allows for the collection, manipulation, and presentation of various data types, resulting in the creation of thematic maps. In recent years, researchers worldwide have utilized multiple approaches to delineate GWPZs. The most commonly used methods include the Analytical Hierarchy Process [5], Frequency Ratio, and Logistic Regression. Moreover, various machine learning techniques are now widely recognized for delineating GWPZs. These methods include Random Forest (RF), Support Vector Machine (SVM), Boosted Regression Trees (BRT), Naïve Bayes, Classification and Regression Trees (CART), and Artificial Neural Networks (ANN) [6].

India is the world's largest consumer of groundwater, using approximately 230 cubic kilometers per year [7]. Approximately 85% of rural households, 50% of urban households, and 65% of irrigation water needs in the country are met by groundwater sources [5]. The 2022 Central Groundwater Board (CGWB) technical report [8] on Jharkhand highlights that the annual extractable groundwater resource in Jharkhand is 5.69 BCM, with 3.92 BCM available for future development. Several areas such as Jamshedpur, Ramgarh, Silli, Ranchi, and Dhanbad shows critical condition. Thus, Groundwater depletion is a critical issue driven by population growth and rising demand that outpaces available water supplies. Therefore, the study aims to map the Groundwater Potential Zones (GWPZs) in the upper Subarnarekha River Basin (SRB), India using Analytical Hierarchy Process (AHP) and Random Forest (RF) model. The SRB consists primarily of hard crystalline rocks, mainly granite gneiss, which are impermeable. As a result, groundwater occurrence is not widespread in the region. Hence, the study's primary objective is to compare the effectiveness of two different approaches, namely the Multi-Criteria Decision-Making (MCDM) method and Machine Learning (ML) in delineating GWPZs within the study area.

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## 2. MATERIALS AND METHODS

### 2.1. Study Area

The upper Subarnarekha River Basin (SRB) spans 14,037 square kilometers across the states of Jharkhand and West Bengal, located between latitudes 22°01' and 23°35' N, and longitudes 85°05' and 86°55' E (Fig. 1). The region receives an annual average rainfall of 1300 to 1800 mm, with the heaviest precipitation occurring from June to October. The yearly average temperature ranges from 23°C to 28°C. The basin is divided into three geological strata: upper and middle parts of the basin consist of pre-Cambrian rocks, and lower part of tertiary and quaternary formations.

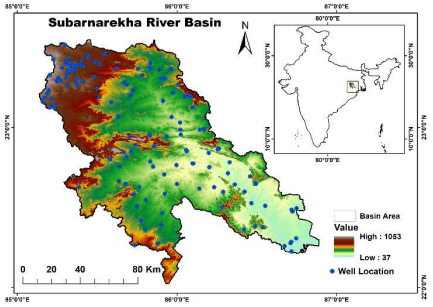



Fig.1 Location Map of the upper Subarnarekha River Basin with observation wells

### 2.2. Dataset

Data from various sources were utilized for spatial modeling and GWPZs analysis (Table 1). All datasets were transformed into a spatial database, and the AHP and RF methods were applied. Fig. 2 illustrates the overall methodological framework of the study.

Table 1 Data Sources

| Data | Source of Data |
|---|---|
| Groundwater level | Central Groundwater Board (CGWB) https://indiawris.gov.in/wris |
| Rainfall | CHIRPS https://www.chc.ucsb.edu/data/chirps |
| Land Use Land Cover (LULC) | Esri Land Cover https://livingatlas.arcgis.com/landcover/ |
| Geology, Geomorphology Lineament | Geological Survey of India, Bhukosh https://bhukosh.gsi.gov.in/Bhukosh/Public |
| Soil map | NBSSLUP https://www.nbsslup.in/ and SAMETI, Jharkhand https://www.sameti.org/district_maps.php |
| DEM (Elevation, Slope, Drainage Density, TWI) | Shuttle Radar Topography Mission (SRTM) https://earthexplorer.usgs.gov/ |

*2.2.1. Well Inventory and Groundwater Conditioning Factors Map*

The inventory map for the study area includes a total of 126 well points, gathered from various sources. A random partition mechanism was used to divide these points into training and testing datasets, with 80% designated for training and 20% for testing.

Selecting effective parameters for GWPZs is a critical task for researchers. The literature review has helped to identify ten key factors that influence groundwater occurrence. It includes Geology, Geomorphology, Elevation, Drainage Density (DD), Lineament Density (LD), Land Use Land Cover (LULC), Rainfall, Soil, Slope, and Topographical Wetness Index (TWI). Based on these parameters, the thematic maps were created using ArcGIS software (Fig. 3). Thematic layers were resampled to a 30 × 30 m grid size.

### 2.3. Methodology

*2.3.1. Analytical Hierarchy Process (AHP)*

AHP is a hierarchical weighting method for multi-criteria decision problems [9]. It evaluates parameters based on their relative importance and ranks them through pairwise comparisons to organize criteria in a hierarchical structure. Each parameter is assigned a set of weights and the data is normalized. Lastly, the Consistency Index (CI) and Consistency Ratio (CR) are estimated to verify the reliability of the assigned weights (Equation 1 and 2).

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (1)$$

where $\lambda_{max}$ the principal eigenvalue of the matrix and n is the total number of criteria.

$$CR = \frac{CI}{RI} \quad (2)$$

where RI is the Random Index which is 1.49 in this case. The CR was determined to be 0.088, which is less than 0.1, indicating that the matrix is consistent.

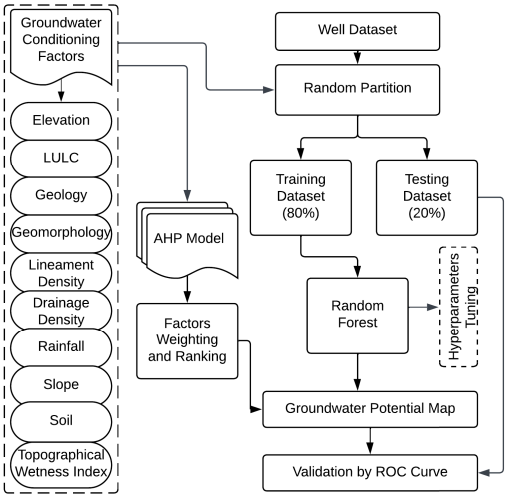



Fig. 2 Methodological Flowchart

*2.3.2. Random Forest (RF)*

Random Forest (RF) is a widely used and highly accurate machine learning algorithm known for its tree-based method.

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It combines multiple decision trees to assess the relationships between factors. Random Forest generates a "forest" of trees using bootstrapped data. It uses 66.66% of the bootstrap data to grow the tree, and the remaining 33.33% to evaluate the tree. The tree's average prediction is calculated using the equation 3.

$$Gp = \frac{1}{k} \sum k^{th}\, v^{response} \tag{3}$$

where *Gp* is the groundwater forecast and k is the number of distinct trees.

The performance of machine learning models heavily depends on dataset variability. Data augmentation techniques like scaling, translation, flipping, and lighting adjustments, commonly used in image recognition, can also enhance spatial datasets [10]. In areas influenced by existing wells, well productivity variation is limited, with wells within the influence zone sharing the same yield class. The well yield data was augmented across neighboring cells (3 × 3 cell radius) to create a dataset of 1134 rows (126 × 9). Each neighboring cell's centroid was assigned the well yield of the original cell, introducing a spatial deviation of 90 m (based on the 30 m cell size). Given the study area's size (14,037 sq. km), this deviation was relatively minor and helped improve results with the small dataset.

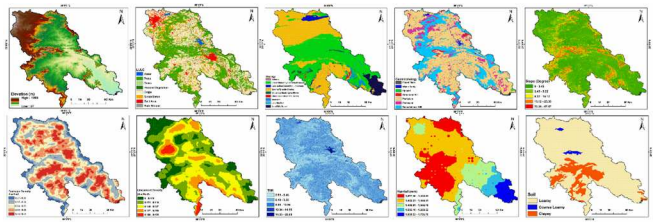


Fig. 3 Thematic layers of groundwater conditioning factors

*2.3.3. Model Validation*

The performance of the groundwater potential model was assessed using accuracy, Kappa index, and ROC curve. The Area Under the Curve (AUC) runs between 0 and 1, with larger values indicating greater model performance. The formula used here are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

$$AUC = \sum TP + \sum TN/(P + N) \tag{5}$$

where TP (True Positives) and TN (True Negatives) represent the number of pixels that are correctly classified. In contrast, FP (False Positives) and FN (False Negatives) represent the number of pixels that are incorrectly classified.

$$Kappa\ (k) = \frac{P_{obs} - P_{exp}}{1 - P_{exp}} \tag{6}$$

Here, $P_{obs}$ represents the number of pixels accurately matched as either well or non-well pixels, while $P_{exp}$ denotes the estimated results.

## 3. RESULTS AND DISCUSSION

### 3.1. Groundwater Potential Zone mapping

Groundwater potential maps for the SRB were generated using AHP and RF shown in Fig. 4. The GWP maps have been classified into five distinct classes based on the Jenk classification scheme: very high, high, moderate, low, and very low.

For the AHP analysis, the thematic layers were classified differently, with the majority being divided into five classes. Based on their relative importance and the opinions of pertinent experts, the criteria and subclasses were sorted and weighted. The model application demonstrated remarkable consistency, with an overall CR of 0.088. After preparing all the raster layers, a weighted overlay analysis was performed to calculate the groundwater potential index. AHP shows satisfactory results with an overall accuracy of 57% and a kappa value of 0.374. Results reveal that approximately 15% of the area has high GWP, primarily in the south and southeast regions. Additionally, around 57% of the area demonstrated moderate GWP in the central part of the basin, whereas 24% of the area showed poor GWP in the northwestern regions. High groundwater potential zones were predominantly located in areas with close proximity to river channels. The alluvial plains in the southern part of the basin showed abundant water-bearing aquifers. Notably, the weighted overlay analysis distinctly highlighted water bodies, providing clear visibility in the delineation. The results indicate that abundant groundwater resources are concentrated near the river channel in areas characterized by a gentle slope, low drainage density, elevated soil moisture and high TWI around the riverbank. Singha et al. [11] discovered that the downstream and southern areas of the region have great groundwater potential, mostly due to the existence of alluvial soil with a high infiltration rate, particularly in flat terrains. In general, southern areas benefit from higher groundwater recharge because runoff moves more slowly than in northern regions.

Initially, the RF model was trained on a dataset of 126 well records. The limited dataset size led to suboptimal performance, with an overall accuracy of 48.09% and kappa index of – 0.032. Therefore, data augmentation techniques were applied to increase the training dataset, aiming to enhance the performance of the machine learning model. The accuracy achieved was 90.97% with a kappa value of 0.820 which is significantly higher than the 126 well data. According to the RF results, 25.51% of the total area is classified as low, 26.97% as moderate, and 21.02% as high GWPZs. The study also concludes that Rainfall, DD, LD, and elevation play a more significant role in influencing GWPZs. The RF model identified the northern and southeastern parts, particularly along the stream networks, as regions with high GWPZs (Fig. 4). In the western region, the extensively dissected plateau was classified as a low potential zone.

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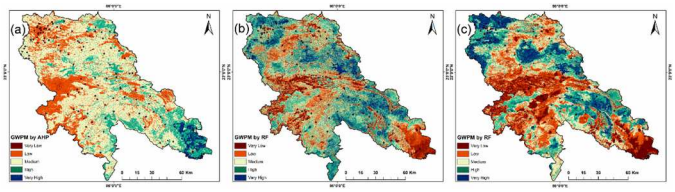


Fig. 4 Groundwater Potential Map derived from (a) AHP, (b) RF (126 well data) and (c) RF (Augmented data)

### 3.2. Validation of the Groundwater Potential maps using ROC curve

The model validation is vital for assessing the GWP maps. The AHP and RF based GWP maps were evaluated using overall accuracy, kappa score and ROC curve (Fig 5). AHP performed satisfactorily with an AUC value of 0.712. The performance of the ML model using the available dataset (126 well points) was found to be unsatisfactory with an AUC value of 0.736. Although the performance of the ML model improved with data augmentation with an AUC value of 0.985, it also introduced a standard deviation equivalent to the neighborhood radius. In the future, the results from groundwater models can be combined with GWPZs to enhance aquifer mapping accuracy.

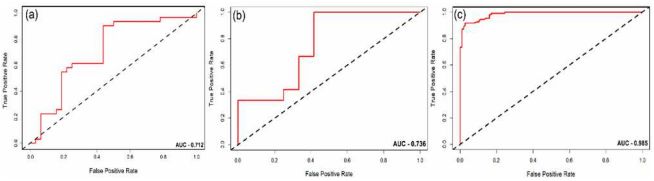


Fig. 5 ROC curve by (a) AHP, (b) RF (126 well data) and (c) RF (Augmented data)

## 5. CONCLUSION

This study presented an innovative and extremely effective approach for estimating groundwater resources in the SRB, India, utilizing AHP and RF models. On average, the models indicated that around 18% of the basin area has a very high potential for groundwater occurrence, with this potential being primarily concentrated in the southern and northern regions of the basin. In contrast, most of the central part of the basin has medium to low potential, which may be due to the higher elevation and geological condition which is mostly composed of Precambrian granitic batholiths. The machine learning model shows the importance of groundwater conditioning factors such as that Rainfall, DD, LD, and elevation in controlling groundwater recharge of the SRB. These findings offer valuable insights for responsible departments to improve groundwater distribution, planning, consumption, and artificial recharge.

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